MIL-B-4173B(USAF)
6 February 1969
SUPERSEDING
MIL-B-4173A(USAF)
18 January 1963

MILITARY SPECIFICATION

BENCHES, WORK, GENERAL PURPOSE

1. SCOPE

*1.1 This specification covers work benches for general maintenance, laboratory, and shop usage.

*1.2 Classification. Work benches furnished under this specification shall be of the following types and classes as specified (see 6.2).

Type I - Work Bench with Drawer Unit (see Fig. 1).
Type II - Work Bench with Tier Drawer Unit (see Fig. 2 and 4).
Type III - Bench Instrument Test (see Fig. 3).

Class 1 - Laminated hardwood top.
Class 2 - Steel top.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

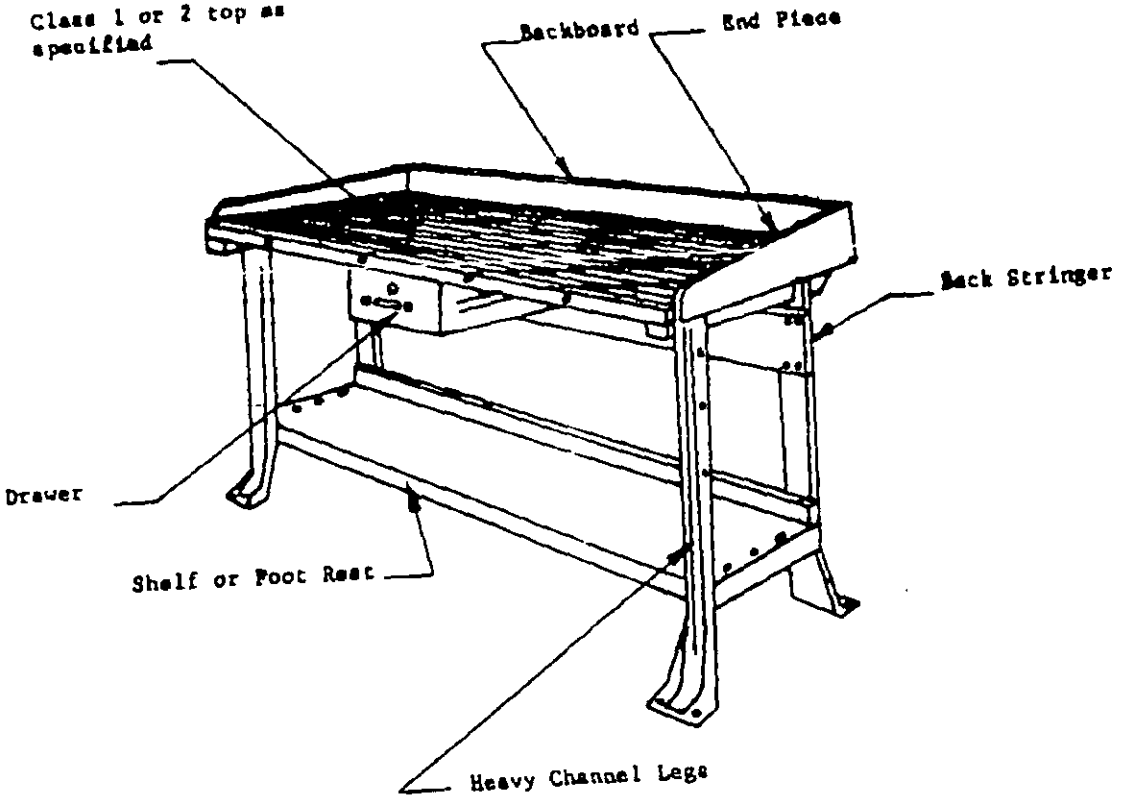
Federal

MM-L-736	Lumber and Timber, Hardwood
TT-V-121	Varnish; Spar, Water Resisting
MMM-A-188	Adhesive, Urea-Resin-Type (Liquid and Powder)
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

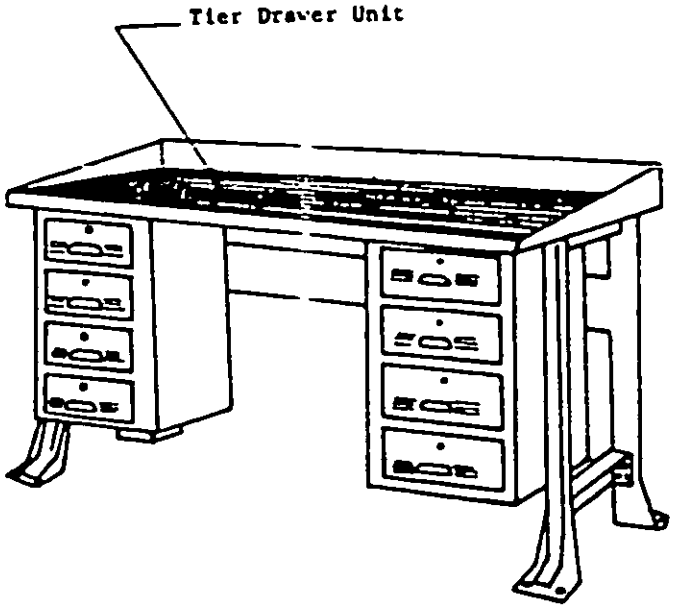
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:26Z
Check the source to verify that this is the current version before use.



Type I Work Bench
FIGURE I

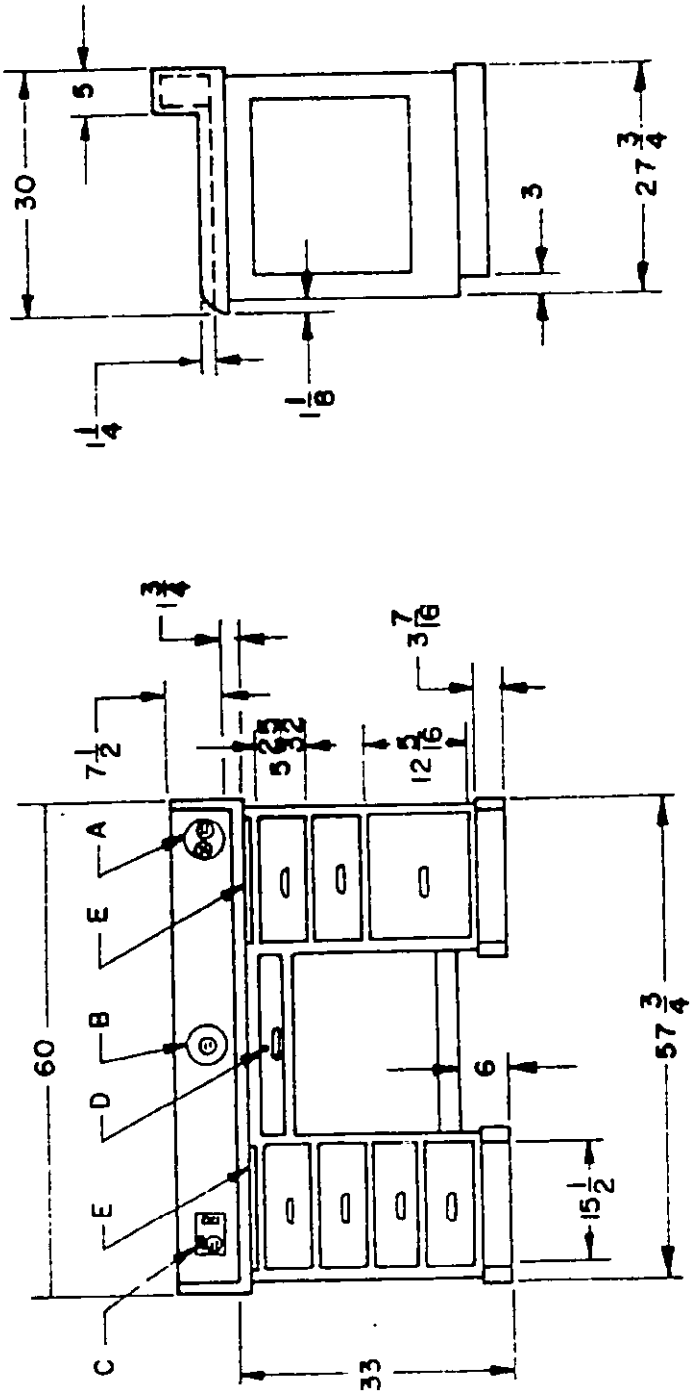
Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:26Z
Check the source to verify that this is the current version before use.



Type II Work Bench

Figure 2

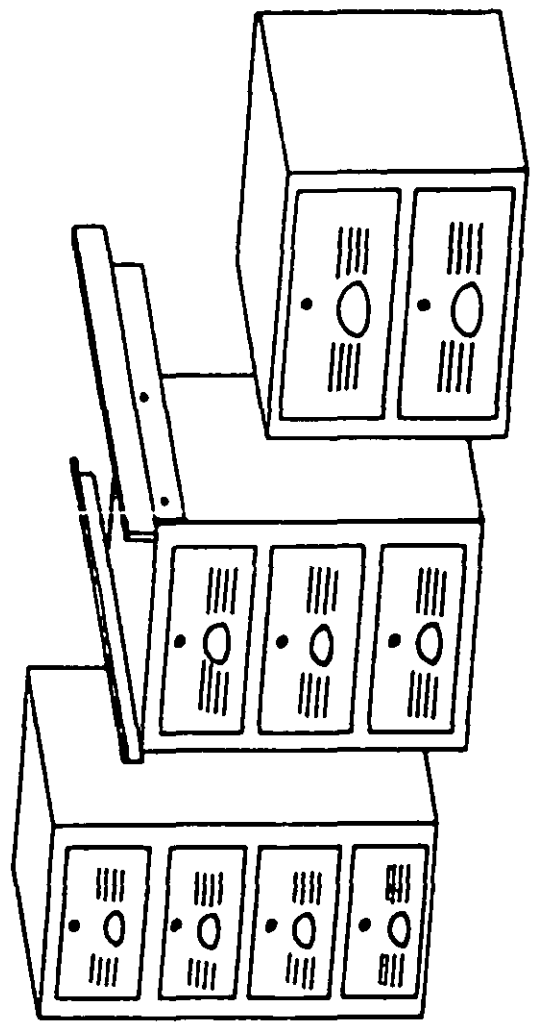
Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:26Z
Check the source to verify that this is the current version before use.



Type III, Instrument Test Bench

Figure 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:26Z
Check the source to verify that this is the current version before use.



Tier Drawer Units

Figure 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:26Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD 595	Colors

Military

MIL-STD-100	Engineering Drawing Practices.
MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Specifications and Standards, Order of Precedence For Selection of.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143.

*3.2 Material. Materials used in construction shall be of such quality as to assure conformance to requirements specified herein.

*3.3 Design and construction. Work benches covered by this specification shall be designed for durability and strength. The design shall be such as to permit easy disassembly and reassembly, without the use of special tools. The benches shall be of all steel construction throughout, except the top section, which may be furnished in two classes.

*3.4 Welding. Welding of metal parts shall be of a type most suitable for the intended purpose and shall be performed in accordance with the best recognized commercial practice. In no case shall welding be resorted to as a repair measure.

*3.5 Type I, Work bench. The type I work bench shall be furnished in 33 ± 1 inch heights, 29 ± 1 inch widths, and in lengths of 5, 6, 8, 10 and 12 feet (see 6.2). They shall consist of an all metal frame structure including leg sections, a reinforced back stringer, lower shelf, drawer with lock, backboard, end piece and bench top. Sizes 5 and 6 foot lengths shall be provided with two leg sections and one drawer located approximately in the center of the table. Sizes 8, 10 and 12 foot lengths shall be provided with three leg sections and two drawers. Each drawer shall be centrally located between the middle and end leg sections. Design and construction shall be substantially in accordance with Figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:26Z
Check the source to verify that this is the current version before use.

*3.5.1 Leg section. The leg section shall consist of two uprights made of heavy formed steel channels with a cross member welded between the uprights approximately 9 inches from the bottom and a rolled pressed steel channel or tie section at least 3 inches wide and not less than 1-3/8 inches deep, welded to the top of uprights. Uprights and cross members shall be of not less than 14 gage U.S.S. (.0781) inches metal. Leg top section shall not be less than 11 gage U.S.S. (.1250) inches metal. Lower portion of the uprights shall be flared to form an integral foot approximately 2-1/2 inches by 6 inches or otherwise provided with a foot not less than 11 gage U.S.S. (.1250) welded to bottom of uprights approximately 3-3/4 inches by 5-1/2 inches. Uprights, cross member, and top section shall be provided with holes for attachment of top, stringers, and shelf or foot rest as required. Foot shall have holes for attaching uprights to floor.

*3.5.2 Back stringer. The back stringer shall be of a heavy formed steel channel or ribbed section of sufficient strength and rigidity to prevent end sway in the assembled bench frame. Channel section shall be made of not less than 14 gage U.S.S (.0781) metal. Stringer shall be provided with necessary holes for attachment to bench legs. A stringer shall be furnished with each bench unit.

*3.5.3 Shelf. Unless otherwise specified, an all metal shelf for storage or foot rest shall be provided (see 6.2). The shelf shall be flanged in front and back for strength, and holes provided for attachment to lower cross member of the bench legs. The shelf shall be of not less than 14 gage U.S.S. (.0781) metal and not less than 16 inches wide. The shelf shall be installed with the front flange down and the back flange either up or down.

*3.5.4 Bench drawer. Each Type I work bench shall be provided with steel drawers 20 ± 1/2 inches long by 20 ± 1/2 inches wide by 6 ± 1/2 inches deep, of not less than 18 gage U.S.S. (.0500) inch. The drawers shall be of welded construction with closed top. The closed top should be enclosed with a tier unit or drawer mounting cover. Runners shall be welded to the cover of each drawer to assure alignment and easy assembly to the work bench. When assembled to the work bench, the drawer shall slide free without binding. The drawer handles shall be designed to provide a comfortable hand hold. Drawers and runners shall be provided with ball bearing rollers or nylon rollers for ease of operation. The quantity of drawers required shall be as specified in paragraph 3.5.

*3.5.4.1 Lock. Each bench drawer shall be provided with a lock of good quality. Two individual keys shall be provided with each bench drawer.

*3.5.5 Backboard and end piece. Unless otherwise specified, an all metal backboard and end piece shall be provided (see 6.2). The backboard shall extend a minimum of 2-1/8 inches above the work surface of the bench and end stops shall taper to approximately 1-1/4 inches at front of bench top

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:26Z
Check the source to verify that this is the current version before use.

or may be a straight 2-1/8" piece. The backboard and end piece shall be flanged for strength and not less than 14 gage U.S.S. (.0781) metal and all necessary holes for attachment to bench top shall be provided.

*3.6 Tops. The 4, 5 and 6 foot tops shall be furnished in one section and the 8, 10 and 12 foot tops may be furnished in two sections.

*3.6.1 Class 1 top. The Class 1 bench top shall be laminated edge grain full length wood strips, 2 inches maximum width, 1-3/4 inches finish thickness. Strips shall be fabricated from good grade hardwood conforming to applicable requirements of MM-L-736, uniformly kiln dried to not over 8 percent moisture content at the time of working. The wood shall be free of decay, splits, wane, warp and unsound knots. The top surface shall be completely free of knots; the bottom surface shall be free of knots exceeding one inch in average diameter, knot clusters, holes or surface cracks. The bench top shall be fabricated with an interlocking joint, either serrated or tongue and grooved. A flat butt joint may be used provided the gluing process produces a quality glue bond. The gluing process may be performed either by curing at room temperature, application of heat from steam heated platens or in a high frequency field. The minimum curing temperature shall be 70°F. The gluing described herein for interlocking joints shall be in accordance with the applicable requirements of MMM-A-188. Bench tops shall be furnished either with or without tie rods. Bench tops provided with tie rods shall be bored from edge to edge to take 3/8 inch tie rods secured with plain nuts. The tie rods shall be located not less than 4 inches from each end of the top and at intervals of approximately 24 inches to 36 inches. The tie rod holes shall be plugged with a flat grain wood plug after assembly. The bench tops furnished without the tie rods shall be made of wood which has been properly dried, machined, and glued so as to guarantee a quality top. All bench tops shall be smooth, flat and uniform not to exceed 1/16 inch in a true plane over the entire top surface.

*3.6.2 Class 2 top. The Class 2 top shall be of heavy guage steel, single sheet not less than No. 12 gage U.S.S. (.1094) inches in thickness, and shall be fabricated with deep box flanges, front and rear for maximum strength. The steel top shall have a smooth working surface and shall be flat and uniform in thickness.

*3.7 Type II work bench. The Type II work bench shall be furnished in 33 ± 1 inch heights, 29 ± 1 inch widths, and in lengths of 4, 5 and 6 foot (see 6.2). The 4 foot length shall contain one tier drawer unit. The 5 and 6 foot length shall contain two tier drawer units. The number of drawers in the tier drawer unit shall be as specified (see 6.2). The Type II work bench shall consist of one Type I frame structure and tier drawers as specified. Design and construction shall be substantially in accordance with Figure 2.

Source: https://assist.dla.mil -- Downloaded: 202640-17T03:26Z
Check the source to verify that this is the currentversion before use.

*3.7.1 Tier drawer unit. Each Type II work bench shall be provided with multiple tier drawer units as specified. The tier drawer units shall be strong and rigidly constructed of steel not less than 20 gage U.S.S. (.0375) inch. The steel drawer shall be 20 ± 1/2 inches long by 20 ± 1/2 inches wide by 6 ± 1/2 inches deep. Each drawer shall be inclosed in an individual cabinet and shall have steel runners provided with ball bearing rollers for assembling two or more units in tiers. Tier units shall be securely mounted to bench tops. Design and construction shall be substantially in accordance with Figure 4.

*3.7.1.1 Lock. A cylinder lock of good quality shall be provided in each drawer, and two individual keys shall be furnished with each single drawer or with each assembled tier drawer unit.

*3.8 Type III bench, instrument test. The Type III benches shall be of wood construction similar in size and design to Figure 3.

*3.8.1 Top. The top for Type III benches shall be as described in paragraph 3.6.1

*3.8.2 Material. The bench frame structure and drawer front shall be made of beech, birch, or maple. All paneling shall be a minimum of 3/16 inches thick, 3 ply, commercial grade A birch or maple plywood. Glue used in the construction shall be of the best commercial grade, moisture resistant type, meeting the requirements of MMM-A-188.

*3.8.3 Drawers. All drawers shall be equipped with desk type drawer pulls. The drawers shall be so designed that the seven side drawers will be automatically locked when the center drawer is fully closed. The center drawer shall have a substantial lock, marked D in Figure 3, provided with two keys. Two desk type slides, marked E. in Figure 3, shall be incorporated in the design with finger pull recesses on the underside.

*3.8.4 Electrical outlets. A panel with removable back for electrical outlets shall be mounted on the top rear of the bench. The panel shall be equipped with one double outlet box, on single outlet box, and one single outlet box with pilot light and toggle switch, marked A, B, and C, respectively in Figure 3. The receptacles shall be individually connected to exposed terminals mounted on the rear panel. All electrical equipment shall be designed for 110 volt.

*3.9 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern manufacturer's part numbers.

*3.10 Identification of product. Work benches and parts shall be marked for identification in accordance with MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:26Z
Check the source to verify that this is the current version before use.

*3.11 Finishing and painting. The equipment shall be finished and painted in accordance with the manufacturer's standard practice for this type of equipment except that a "crackle" or "wrinkle" finish will not be acceptable. The wooden parts shall be given 2 coats of spar varnish conforming to TT-V-121. Unless otherwise specified, clear varnish shall be used on wood surfaces and the color of other surfaces, except electrical furnishings, shall be Aircraft Gray, Color No. 16473 of FED-STD-595.

*3.12 Workmanship. Workmanship shall be in accordance with high-grade commercial practice to assure a quality product. Components shall be properly assembled as well as having a neat general appearance.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Classification of tests. All the tests required for the testing of core material are classified as quality conformance tests.

4.2 Sampling.

*4.2.1 Lot. Unless otherwise specified in the contract or order, a lot shall consist of all benches manufactured under the same conditions and submitted for acceptance at the same time.

*4.2.2 Sampling plans. Unless otherwise specified in the contract, purchase order or drawing information sampling plans and procedures in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions set forth in MIL-STD-105.

*4.2.2.1 Visual and dimensional examination. A random sample of benches shall be selected from each lot in accordance with procedures of MIL-STD-105, Inspection Level II. The Acceptable Quality Level shall be 1.5 percent defective for major defects and 6.5 percent defective for minor defects.

*4.3 Examination. Each of the sample benches selected in accordance with 4.2.2.1 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in Table I. Any bench in the sample containing one or more defects shall be rejected and if the number of defective benches in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 202610-17T03:26Z
Check the source to verify that this is the current version before use.

TABLE I - CLASSIFICATION OF DEFECTS

Categories	Defects
Major	(AQL shall be 1.5 percent defective)
101	Defective welding, brazing and soldering affecting serviceability
102	Defective or insecure fastening devices affecting serviceability
103	Benches not complete, component parts missing
104	Overall dimensions not as specified
105	Material not as specified
106	Design not as specified
Minor	(AQL) shall be 6.5 percent defective)
201	Painting and finishes not as specified
202	Drawers handles and locks not as specified
203	Component parts not new; evidence of reclaimed, used, or salvaged components
204	Wood top tie rods not as specified
205	Identification marking not as specified
206	Preservation and packaging not as specified

4.3 Inspection and preparation for delivery. The preservation, packaging and marking of the work benches shall be examined to determine conformance with the requirements of Section 5 of this specification, and as required by specifications referenced therein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with MIL-STD-105 with inspection Level II and an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Packaging. (see 6.2).

5.1.1 Level A. Each bench shall be preserved and packaged in accordance with MIL-P-116, Method III.

5.1.2 Level C. Each bench shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:26Z
Check the source to verify that this is the current version before use.

*5.2.1 Level A. Each bench packaged as specified in 5.1 shall be disassembled and individually unit packed complete with all parts necessary for assembly at destination. Small parts, such as bolts, nuts, washers, etc., unless otherwise protected by a suitable rust preventative, shall be bagged in greaseproof material conforming to MIL-B-121, all parts shall be blocked or braced within overseas type containers conforming to PPP-B-621. Containers shall be provided with case liners conforming to MIL-L-10547. Containers shall be closed and strapped in accordance with applicable specification and appendix thereto.

*5.2.2 Level B. Each bench shall be packed same as specified in 5.1.1 except that containers shall be of Domestic type and case liners will not be used.

*5.2.3 Level C. Each bench complete with all parts shall be packed to insure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification Rules or other carrier rules applicable to the mode of transportation.

*5.3 Marking. In addition to any special marking required by the contract or order, containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The work benches covered by this specification are for general maintenance, laboratory, and shop usage.

*6.2 Ordering data.

a. Title, number, and date of this specification.

b. Specify type, class, and size required (see 1.2, 3.5, and 3.7).

c. Specify if metal shelf is required (see 3.5.3).

d. Specify if metal backboard and end piece is required (see 3.5.5).

e. Specify number of drawers required in tier drawer unit (see 3.7).

f. Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

*6.3 Changes from previous issue. The outside margins of this document have been marked with an asterisk to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 202610-17T03:26Z
Check the source to verify that this is the current version before use.

CUSTODIANS
Air Force - 84

PREPARING ACTIVITY
Air Force - 84

PROJECT NUMBER
7195-0055

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:26Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

WRAMA (WRNEMI)
Robins AFB, Georgia 31093

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-01-17 08:28Z
Check the source to verify that this is the current version before use.